Lctr 17: Inverse Laplace Transform, Pole-Zero Analysis, and Properties

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Focus: Sections 9.3–9.6 (Pages 670–692)

17.1 The Big Picture

Why This Matters

In Lecture 16 we learned how to go *forward*: given a time-domain signal $x(t)$, compute its Laplace transform $X(s)$ and ROC. In practice, we almost always need to go in the *reverse* direction: given a system's transfer function $H(s)$ (which we get from a differential equation), find the corresponding impulse response $h(t)$.
This lecture covers three essential skills: (1) how to invert the Laplace transform using partial fractions, (2) how to read system behavior directly from pole-zero plots without doing algebra, and (3) the properties of the Laplace transform that let us handle operations like differentiation, convolution, and time shifting algebraically.

Roadmap:

1. The inverse Laplace transform via partial fraction expansion.
2. Worked examples: distinct real poles, repeated poles, complex poles.
3. Geometric evaluation of the Fourier transform from pole-zero plots.
4. Properties of the Laplace transform (with ROC implications).
5. Summary table of transform pairs and properties.

17.2 The Inverse Laplace Transform

17.2.1 The Formal Inversion Integral

The exact inverse is:

$$x(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} X(s)\, e^{st}\, ds$$

This is a contour integral in the complex plane along a vertical line in the ROC. In this course, we will **never evaluate this integral directly**. Instead, we use a far more practical method.

17.2.2 The Practical Method: Partial Fraction Expansion

The strategy is:

1. Express $X(s)$ as a sum of simple terms $\frac{A_i}{s-p_i}$, $\frac{A_i}{(s-p_i)^n}$, etc.
2. Invert each term using the table from Lecture 16.
3. Use the ROC to determine whether each term is right-sided or left-sided.

Key Insight

The ROC tells you which direction each exponential goes. For a pole at $s = p_i$:

- If the ROC is to the **right** of p_i: the term is $e^{p_i t}\, u(t)$ (right-sided).
- If the ROC is to the **left** of p_i: the term is $-e^{p_i t}\, u(-t)$ (left-sided).

This is the only place where the ROC enters the inversion process, but it is critical.

17.3 Example: Distinct Real Poles

Inverse Laplace Transform with Distinct Real Poles

Find $x(t)$ given:

$$X(s) = \frac{2s+6}{(s+1)(s+3)}, \qquad \mathrm{Re}\{s\} > -1$$

Step 1: Partial fraction expansion.

$$\frac{2s+6}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

Multiply both sides by $(s+1)(s+3)$:

$$2s + 6 = A(s+3) + B(s+1)$$

Cover-up method:

- Set $s = -1$: $2(-1) + 6 = A(-1+3) \;\Rightarrow\; 4 = 2A \;\Rightarrow\; A = 2$.
- Set $s = -3$: $2(-3) + 6 = B(-3+1) \;\Rightarrow\; 0 = -2B \;\Rightarrow\; B = 0$.

So: $X(s) = \dfrac{2}{s+1} + \dfrac{0}{s+3} = \dfrac{2}{s+1}$

Step 2: Check the ROC for each pole.

The ROC is $\mathrm{Re}\{s\} > -1$. The pole at $s = -1$ has the ROC to its *right*, so the corresponding term is right-sided.

Step 3: Invert.

$$\boxed{x(t) = 2\, e^{-t}\, u(t)}$$

Check: At $t = 0^+$: $x(0^+) = 2$. From the initial value theorem (which we will see later), $\lim_{s\to\infty} sX(s) = \lim_{s\to\infty} \frac{2s^2+6s}{s^2+4s+3} = 2$. ✓

17.4 Example: Distinct Real Poles with Mixed ROC

Two-Sided Signal from Partial Fractions

Find $x(t)$ given:

$$X(s) = \frac{5s+17}{(s+1)(s-3)}, \qquad -1 < \mathrm{Re}\{s\} < 3$$

Step 1: Partial fractions.

$$\frac{5s+17}{(s+1)(s-3)} = \frac{A}{s+1} + \frac{B}{s-3}$$

- $s = -1$: $5(-1) + 17 = A(-1-3) \;\Rightarrow\; 12 = -4A \;\Rightarrow\; A = -3$.
- $s = 3$: $5(3) + 17 = B(3+1) \;\Rightarrow\; 32 = 4B \;\Rightarrow\; B = 8$.

So: $X(s) = \dfrac{-3}{s+1} + \dfrac{8}{s-3}$

Step 2: Determine the direction of each term from the ROC.
The ROC is the strip $-1 < \sigma < 3$.

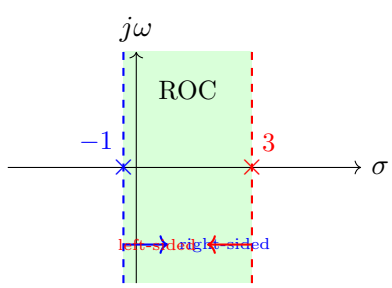
- Pole at $s = -1$: The ROC is to the **right** of this pole $\Rightarrow$ right-sided.
- Pole at $s = 3$: The ROC is to the **left** of this pole $\Rightarrow$ left-sided.

Step 3: Invert each term.

$$\frac{-3}{s+1} \text{ (right-sided)} \longrightarrow -3\,e^{-t}\,u(t)$$

$$\frac{8}{s-3} \text{ (left-sided)} \longrightarrow -8\,e^{3t}\,u(-t)$$

$$\boxed{x(t) = -3\,e^{-t}\,u(t) - 8\,e^{3t}\,u(-t)}$$



17.5 Example: Repeated Poles

Repeated Poles

Find $x(t)$ given:

$$X(s) = \frac{4s+5}{(s+1)^2(s+3)}, \qquad \mathrm{Re}\{s\} > -1$$

Step 1: Partial fraction expansion.
For a repeated pole at $s = -1$ of order 2, we need:

$$\frac{4s+5}{(s+1)^2(s+3)} = \frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s+3}$$

Multiply through by $(s+1)^2(s+3)$:

$$4s+5 = A(s+1)(s+3) + B(s+3) + C(s+1)^2$$

Find C: set $s = -3$: $4(-3)+5 = C(-3+1)^2 = 4C \;\Rightarrow\; C = -7/4$.
Find B: set $s = -1$: $4(-1)+5 = B(-1+3) = 2B \;\Rightarrow\; B = 1/2$.
Find A: compare coefficients of s^2: $0 = A + C \;\Rightarrow\; A = -C = 7/4$.
So:

$$X(s) = \frac{7/4}{s+1} + \frac{1/2}{(s+1)^2} + \frac{-7/4}{s+3}$$

Step 2: ROC is $\mathrm{Re}\{s\} > -1$. All poles are at $\sigma \le -1$, so all terms are right-sided.
Step 3: Invert using the pair $\frac{1}{(s+a)^2} \leftrightarrow t\,e^{-at}\,u(t)$:

$$\boxed{x(t) = \frac{7}{4}\,e^{-t}\,u(t) + \frac{1}{2}\,t\,e^{-t}\,u(t) - \frac{7}{4}\,e^{-3t}\,u(t)}$$

Warning

For repeated poles of order n, the partial fraction expansion includes terms $\frac{B_1}{(s-p)}, \frac{B_2}{(s-p)^2}, \ldots, \frac{B_n}{(s-p)^n}$. The inverse of $\frac{1}{(s-p)^k}$ involves $t^{k-1}\,e^{pt}$, not just e^{pt}. Forgetting the lower-order terms $\frac{B_1}{s-p}, \ldots$ is a very common mistake.

17.6 Example: Complex Conjugate Poles

Complex Poles: Second-Order System

Find $x(t)$ given:

$$X(s) = \frac{2s}{s^2+4s+13}, \qquad \mathrm{Re}\{s\} > -2$$

Step 1: Find the poles by completing the square:

$$s^2+4s+13 = (s+2)^2 + 9 = (s+2)^2 + 3^2$$

Poles are at $s = -2 \pm j3$ (complex conjugate pair with $\sigma_0 = -2$, $\omega_d = 3$).
Step 2: Rewrite the numerator in terms of $(s+2)$:

$$2s = 2(s+2) - 4$$

So:

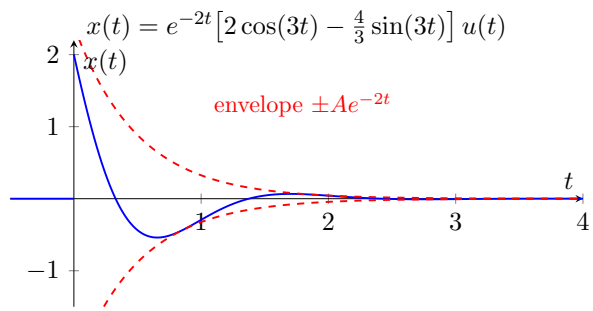
$$X(s) = \frac{2(s+2)}{(s+2)^2+9} - \frac{4}{(s+2)^2+9} = 2 \cdot \frac{(s+2)}{(s+2)^2+3^2} - \frac{4}{3} \cdot \frac{3}{(s+2)^2+3^2}$$

Step 3: Invert using the pairs:

$$\frac{s+a}{(s+a)^2+\omega_d^2} \;\mathcal{L}\; e^{-at}\cos(\omega_d t)\,u(t)$$
$$\frac{\omega_d}{(s+a)^2+\omega_d^2} \;\mathcal{L}\; e^{-at}\sin(\omega_d t)\,u(t)$$

$$\boxed{x(t) = \left[2\,e^{-2t}\cos(3t) - \frac{4}{3}\,e^{-2t}\sin(3t)\right]u(t)}$$

This is a damped sinusoid: the e^{-2t} envelope decays (since the real part of the poles is $-2 < 0$), and the oscillation frequency is $\omega_d = 3$ rad/s.



Key Insight

Complex conjugate poles always come in pairs $s = -\sigma_0 \pm j\omega_d$. They produce **damped sinusoidal** time-domain signals of the form $e^{-\sigma_0 t}\left[A\cos(\omega_d t) + B\sin(\omega_d t)\right]u(t)$. The real part $-\sigma_0$ controls the decay rate; the imaginary part ω_d controls the oscillation frequency. This is the same behavior we saw in second-order systems in Lecture 15, now derived systematically from the Laplace transform.

17.7 Geometric Evaluation from Pole-Zero Plots

17.7.1 The Key Formula

Given $X(s)$ in factored form:

$$X(s) = K\,\frac{\prod_{i=1}^{M}(s-z_i)}{\prod_{i=1}^{N}(s-p_i)}$$

Each factor $(s - z_i)$ or $(s - p_i)$ is a complex number. We can write it in polar form:

$$(s - z_i) = |s - z_i| \, e^{j\angle(s - z_i)}$$

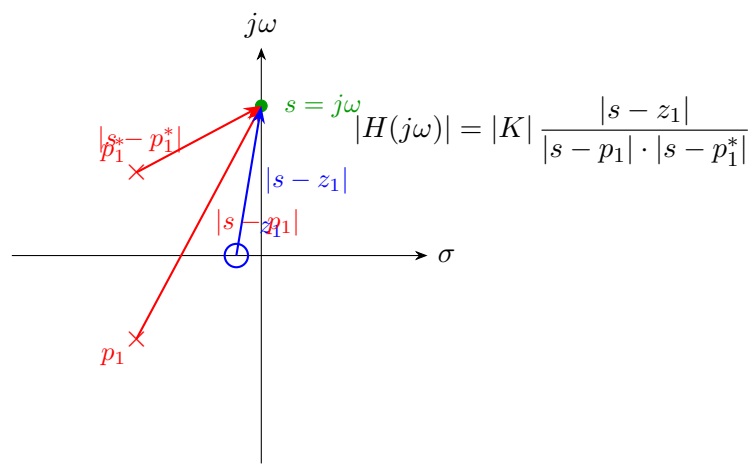
The magnitude and phase of $X(s)$ are therefore:

$$\boxed{|X(s)| = |K| \frac{\prod_{i=1}^{M} |s - z_i|}{\prod_{i=1}^{N} |s - p_i|}} \qquad \boxed{\angle X(s) = \angle K + \sum_{i=1}^{M} \angle(s - z_i) - \sum_{i=1}^{N} \angle(s - p_i)}$$

Geometrically, $|s - p_i|$ is the **distance** from the point s to the pole p_i in the s-plane, and $\angle(s - p_i)$ is the **angle** of the vector from p_i to s.

17.7.2 Evaluating the Frequency Response

To get the frequency response $H(j\omega)$, we evaluate at $s = j\omega$ (points on the $j\omega$-axis):



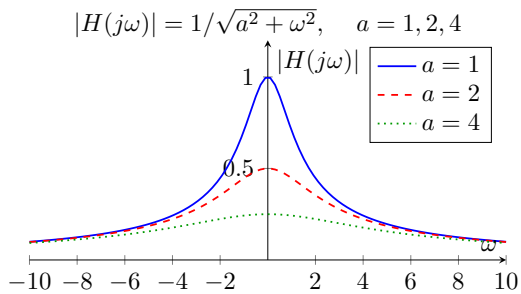
Key Insight

Near a pole: the denominator vector gets short $\Rightarrow$ $|H|$ gets large (resonance peak).
Near a zero: the numerator vector gets short $\Rightarrow$ $|H|$ gets small (notch/null).
This gives you a quick qualitative picture of the frequency response by just looking at the pole-zero plot.

17.7.3 First-Order System Example

Consider $H(s) = \dfrac{1}{s + a}$ with $a > 0$ (single pole at $s = -a$, no finite zeros):

$$|H(j\omega)| = \frac{1}{|j\omega + a|} = \frac{1}{\sqrt{a^2 + \omega^2}}$$



When the pole is farther from the $j\omega$-axis (larger a), the minimum distance from any point on the $j\omega$-axis to the pole is larger, so the peak magnitude is lower and the response is wider. This connects directly to the bandwidth–decay-rate relationship from Lecture 15.

17.8 Properties of the Laplace Transform

In the table below, $x(t) \mathcal{L} X(s)$ with ROC R_x, and $y(t) \mathcal{L} Y(s)$ with ROC R_y.

Property	**Time Domain**	**s-Domain / ROC**
Linearity	$ax(t) + by(t)$	$aX(s) + bY(s)$, ROC $\supseteq R_x \cap R_y$
Time shifting	$x(t - t_0)$	$e^{-st_0}X(s)$, ROC $= R_x$
s-domain shifting	$e^{s_0 t}x(t)$	$X(s - s_0)$, ROC $= R_x$ shifted by $\mathrm{Re}\{s_0\}$
Time scaling	$x(at)$, $a > 0$	$\frac{1}{a} X\left(\frac{s}{a}\right)$, ROC $= aR_x$
Conjugation	$x^*(t)$	$X^*(s^*)$, ROC $= R_x$
Convolution	$x(t) * y(t)$	$X(s) \cdot Y(s)$, ROC $\supseteq R_x \cap R_y$
Differentiation in t	$\frac{d}{dt}x(t)$	$sX(s)$, ROC $\supseteq R_x$
Integration in t	$\int_{-\infty}^{t} x(\tau)\, d\tau$	$\frac{1}{s}X(s)$, ROC $\supseteq R_x \cap \{\mathrm{Re}\{s\} > 0\}$
Differentiation in s	$-t\, x(t)$	$\frac{d}{ds}X(s)$, ROC $= R_x$

17.8.1 Why These Properties Matter

Key Insight

Convolution $\longleftrightarrow$ Multiplication: This is the most important property for system analysis. If a system has impulse response $h(t) \leftrightarrow H(s)$ and input $x(t) \leftrightarrow X(s)$, then the output is:

$$y(t) = x(t) * h(t) \;\; \mathcal{L} \;\; Y(s) = X(s) \cdot H(s)$$

Convolution in time becomes multiplication in s, the same simplification as the Fourier transform, but now applicable to a much wider class of signals.

Differentiation $\longleftrightarrow$ Multiplication by s: This is why the Laplace transform turns differential equations into algebraic equations. Each derivative d/dt becomes a multiplication by s.

17.8.2 Worked Example: Using the s-Domain Shift Property

Using s-Domain Shifting

Find the Laplace transform of $x(t) = e^{-3t}\cos(5t)\,u(t)$.

Step 1: Start with the known pair:

$$\cos(5t)\,u(t) \;\; \mathcal{L} \;\; \frac{s}{s^2+25}, \quad \mathrm{Re}\{s\} > 0$$

Step 2: Apply the s-domain shift: multiplication by e^{-3t} in time $\Rightarrow$ replace s by $s+3$ in the transform:

$$e^{-3t}\cos(5t)\,u(t) \;\; \mathcal{L} \;\; \frac{s+3}{(s+3)^2+25}, \quad \mathrm{Re}\{s\} > -3$$

Notice the ROC shifted left by 3: from $\sigma > 0$ to $\sigma > -3$.

17.8.3 Worked Example: Using the Differentiation Property

Transform of $t\,e^{-2t}\,u(t)$ via the Differentiation-in-s Property

Step 1: Start with:

$$e^{-2t}\,u(t) \;\mathcal{L}\; \frac{1}{s+2}, \quad \mathrm{Re}\{s\} > -2$$

Step 2: Apply the property $-t\,x(t) \leftrightarrow \frac{d}{ds}X(s)$:

$$-t\,e^{-2t}\,u(t) \;\mathcal{L}\; \frac{d}{ds}\left(\frac{1}{s+2}\right) = \frac{-1}{(s+2)^2}$$

Step 3: Multiply both sides by -1:

$$\boxed{t\,e^{-2t}\,u(t) \;\mathcal{L}\; \frac{1}{(s+2)^2}, \quad \mathrm{Re}\{s\} > -2}$$

This confirms the repeated-pole pair from the table.

17.9 The Initial-Value and Final-Value Theorems

These theorems let you extract the behavior of $x(t)$ at $t = 0^+$ and $t \to \infty$ directly from $X(s)$, without inverting.

Initial-Value Theorem

If $x(t) = 0$ for $t < 0$ and $x(t)$ contains no impulses at $t = 0$:

$$\boxed{x(0^+) = \lim_{s\to\infty} s\,X(s)}$$

Final-Value Theorem

If $x(t) = 0$ for $t < 0$ and $x(t)$ has a finite limit as $t \to \infty$ (i.e., all poles of $sX(s)$ have negative real parts):

$$\boxed{x(\infty) = \lim_{s\to 0} s\,X(s)}$$

Warning

The final-value theorem is only valid if $\lim_{t\to\infty} x(t)$ actually exists as a finite number. If $x(t)$ oscillates (poles on the $j\omega$-axis) or grows (poles in the RHP), the theorem gives a **meaningless result**. Always check that $sX(s)$ has all its poles in the open left half-plane before applying it.

Initial and Final Value Theorems

Let $X(s) = \dfrac{10}{s(s+2)(s+5)}$, ROC: $\mathrm{Re}\{s\} > 0$.

Initial value:

$$x(0^+) = \lim_{s\to\infty} s \cdot \frac{10}{s(s+2)(s+5)} = \lim_{s\to\infty} \frac{10}{(s+2)(s+5)} = 0$$

Final value: Check: $sX(s) = \frac{10}{(s+2)(s+5)}$ has poles at $s = -2, -5$ (both in LHP). ✓

$$x(\infty) = \lim_{s\to 0} \frac{10}{(s+2)(s+5)} = \frac{10}{(2)(5)} = 1$$

So $x(t)$ starts at 0 and settles to 1 as $t \to \infty$, a step response that rises from zero to a final value of 1.

17.10 Comprehensive Laplace Transform Pairs

Signal $x(t)$	$X(s)$	**ROC**
$\delta(t)$	1	all s
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}\{s\} > 0$
$t\,u(t)$	$\frac{1}{s^2}$	$\mathrm{Re}\{s\} > 0$
$\frac{t^{n-1}}{(n-1)!}\,u(t)$	$\frac{1}{s^n}$	$\mathrm{Re}\{s\} > 0$
$e^{-at}\,u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}\{s\} > -\mathrm{Re}\{a\}$
$t\,e^{-at}\,u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}\{s\} > -\mathrm{Re}\{a\}$
$e^{-at}\cos(\omega_d t)\,u(t)$	$\frac{s+a}{(s+a)^2+\omega_d^2}$	$\mathrm{Re}\{s\} > -a$
$e^{-at}\sin(\omega_d t)\,u(t)$	$\frac{\omega_d}{(s+a)^2+\omega_d^2}$	$\mathrm{Re}\{s\} > -a$

17.11 Summary

1. **Partial fractions** is the practical method for inverting Laplace transforms: decompose $X(s)$ into simple terms, invert each using the table, and use the ROC to determine right-sided vs. left-sided.

2. **Distinct real poles** give pure exponentials. **Repeated poles** give polynomial-times-exponential terms. **Complex poles** give damped sinusoids.

3. The **pole-zero plot** gives geometric insight: the frequency response magnitude is the product of zero-distances divided by the product of pole-distances. Near a pole ⇒ large magnitude (resonance). Near a zero ⇒ small magnitude (notch).

4. The **convolution property** ($x * h \leftrightarrow XH$) is the foundation of system analysis. The **differentiation property** ($dx/dt \leftrightarrow sX$) turns differential equations into algebra.

5. The **initial-** and **final-value theorems** let you check boundary behavior without full inversion.

17.12 Common Mistakes to Avoid

1. **Forgetting to use the ROC to assign direction:** Each term in the partial fraction expansion can be right-sided or left-sided. The ROC, not the sign of the pole, determines which.

2. **Incomplete partial fractions for repeated poles:** A pole of order n at $s = p$ requires n terms: $\frac{B_1}{s-p}, \frac{B_2}{(s-p)^2}, \ldots, \frac{B_n}{(s-p)^n}$.

3. **Wrong handling of complex poles:** Never split complex conjugate poles into separate $\frac{A}{s-p}$ and $\frac{A^*}{s-p^*}$ terms and try to invert them individually with real exponentials. Instead, complete the square and use the cos/sin pairs.

4. **Applying the final-value theorem when it does not apply:** If $sX(s)$ has poles on the $j\omega$-axis or in the RHP, the theorem is invalid. Always check first.

5. **Confusing pole-zero distances with pole-zero locations:** The frequency response depends on *distances from $s = j\omega$ to each pole/zero*, not on the pole/zero locations themselves.

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